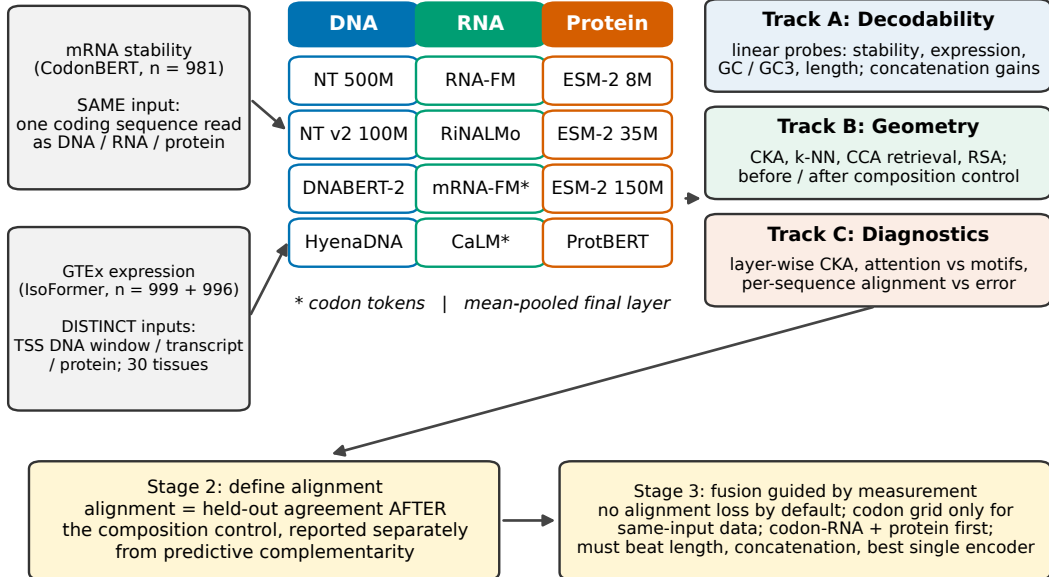


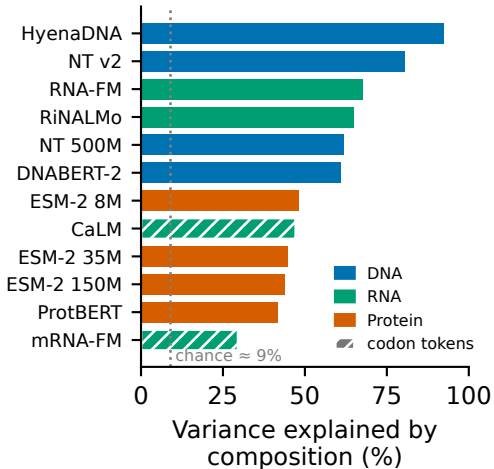
Data

12 frozen encoders

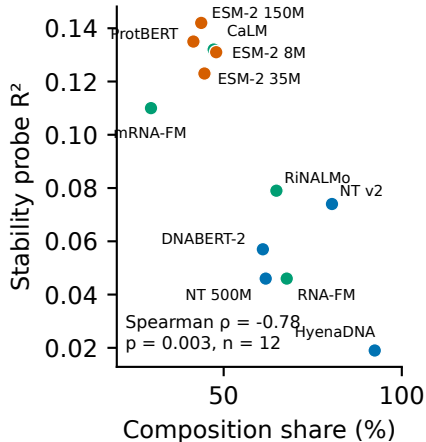
Stage 1 analyses



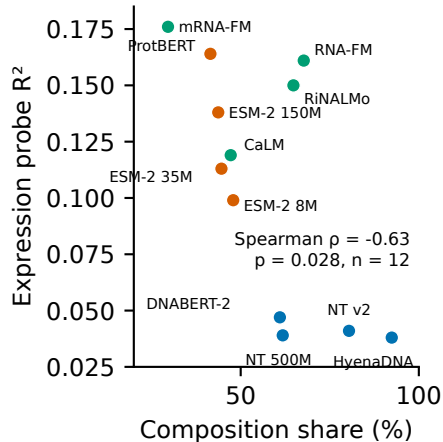
a Composition share



b mRNA stability

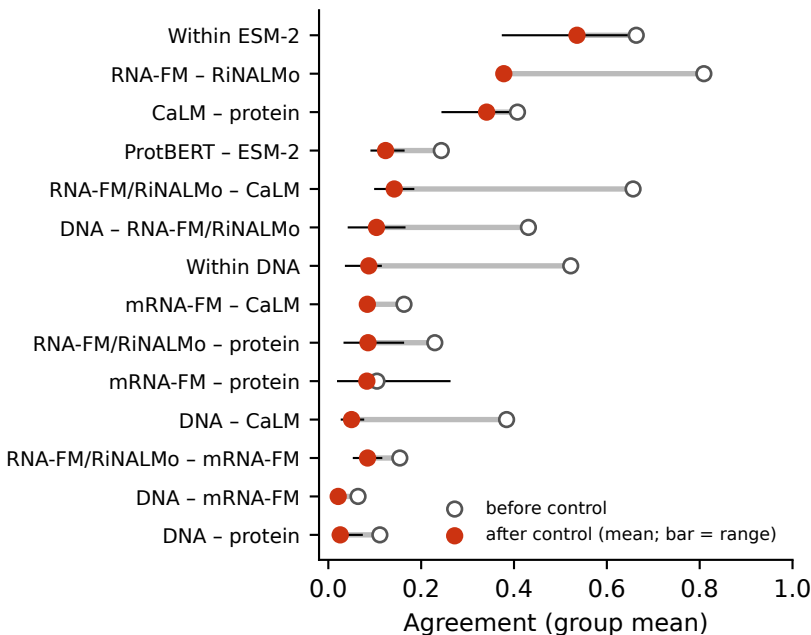


c GTEx expression

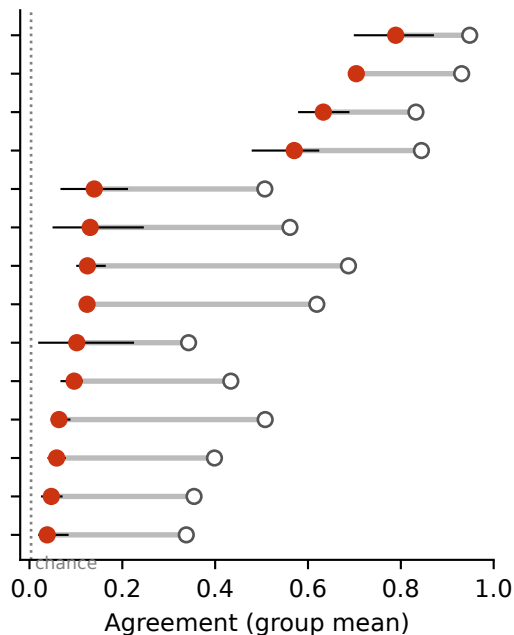


Agreement between encoders before and after removing composition (stability data, 66 pairs)

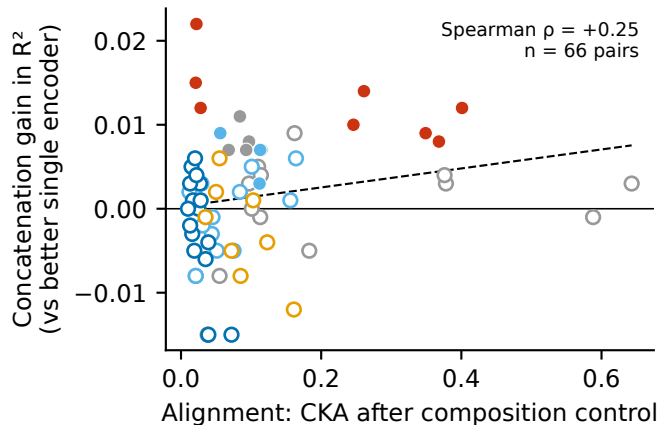
a Linear CKA



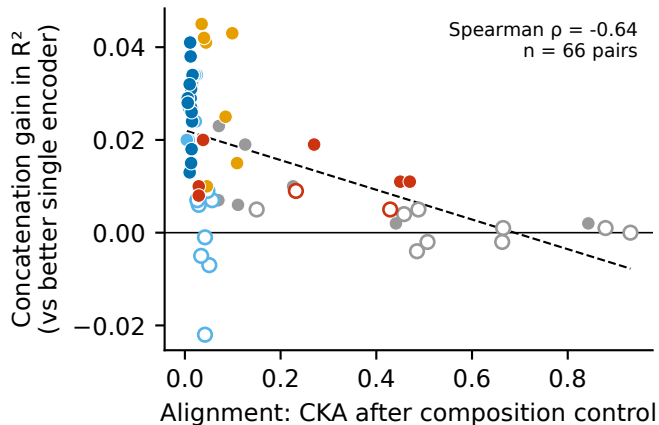
b Retrieval Recall@1



a mRNA stability: same input



b GTEx: distinct inputs



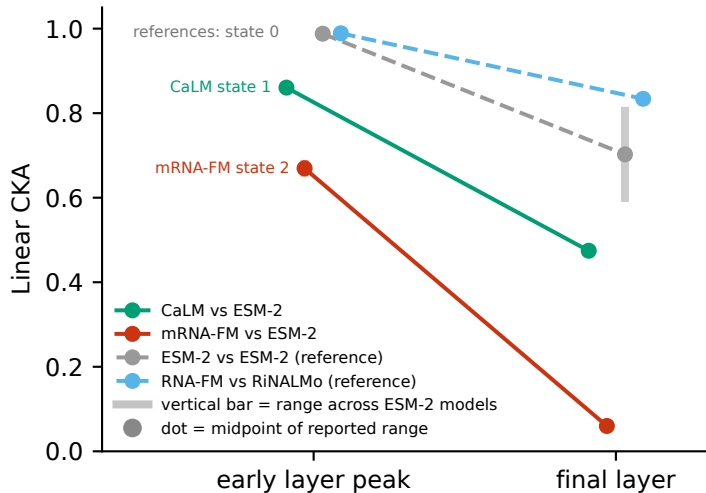
Colour: pair type

- within modality
- DNA - RNA
- DNA - protein
- nucleotide RNA - protein
- codon RNA - protein

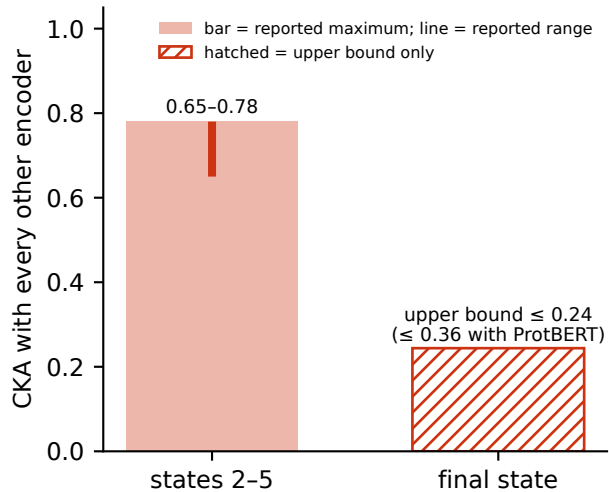
Marker

- filled = gain under all 10 fold assignments (consistent)
- open = gain under fewer than 10 assignments (inconsistent)
- least-squares trend

a Where agreement lives



b mRNA-FM diverges in its last blocks



Values as reported in Section 2.4.3 (194-sequence subsample; ranges span the ESM-2 models). Full layer $\times$ layer heatmaps: run fig5_heatmap_template.py on the notebook's CKA matrices.